Data Cleaning Report

AeroGrid | Source: live Neon warehouse | UTC: 2026-11-13T13:41:53.742811+00:00

Run details

Latest ETL run ID: 7, mode=validate, started=2026-11-13 13:19:02.457585+00:00, status=SUCCESS, version=wp08-v1

Staging and warehouse counts

Component	Stored records
Staging ADS-B positions (all statuses)	27479
Warehouse tracking facts	27479
Stored rejection records	418
Staged FAA airspace features	0
Staged FAA restriction snapshots	0

Source inventory and provenance (latest 25)

Source	License	Fetched UTC	Bytes	SHA-256
TARTAN	CC BY 4.0	2026-11-11 03:51:52.949066+00:00	4325376	86342d09e5004102d765b82d04d16b43b6f15f2e7d829bfbde7909187d08a762

Source URLs and attribution

Source 1: TARTAN | URL: https://airlab-cloud.andrew.cmu.edu:8080/swift/v1/AUTH_ac8533a83cff4d48bc8c608ad222d330/tartanaviation-adsb/kbtp/raw/2022.zip | Attribution: Patrikar et al., Scientific Data 12:468 (2025), DOI 10.1038/s41597-025-04775-6; TartanAviation | Object key: samples/kbtp/01-01-22/1.csv

Rejected records by source and rule

Source	Reason	Count
TARTAN	INVALID_VEHICLE_ID	416
TARTAN	INVALID_TIMESTAMP	2

Cleaning methods

Source normalizers validate timestamps, position units, identifiers, and coordinates. Accepted records enter staging; rejected records retain an auditable reason. Staging rows include all statuses, so staging totals must not be called accepted totals. Source-specific duplicate handling is recorded by ETL run.

Limitations and provenance

January 2022 KBTP Tartan positions are historical, while current FAA TFR snapshots are from 2026. Do not join present restrictions to historical flights. Unknown on_ground values remain NULL; airborne eligibility is inferred in WP07-AIR-01. The spatial grid is 2D (0.01-degree), not an altitude-band grid.

Data authenticity

Production evidence is based on source-linked real rows, not synthetic demonstration records. Zero-source coverage is reported as zero, not imputed.

Reproduction

Run WP-04 staging checks, WP-06 core fact reconciliation and WP-07 derived facts verifier through protected warehouse CI. Regenerate this PDF using python -m etl.reports with the read-only Neon connection in CI.